

Figure 6.2 -- Two independent vacuity channels: mechanism, silence, fix, evidence

Channel 1
a complete model checker that could not fail

Mechanism

lifter.cpp never calls set_buchi /
set_acceptance / set_generalized_buchi.
A non-looping automaton has an
empty omega-language -> vacuous
COMPLIANT on every property.

Why tests were silent

The passing test suite documented
vacuity as intended semantics:
assert result.verdict == "COMPLIANT"
vacuously true (pre-fix,
test_cpp_engine.py:407).

Fix

Commit 37a3d81: negate spot::from_ltlf
(phi, "alive") against an alive-extended
copy when the automaton has no cycle
(0/43 eligible corpus graphs do).

Evidence

G(!B) on a WIR where B provably
executes: COMPLIANT pre-fix,
VIOLATION post-fix (real build).

Channel 2 -- disjoint atom vocabularies

Mechanism

Spec atoms carry start_/done_
lifecycle prefixes; the lifter emits
bare task names. 0 of 116 spec
atoms can ever match.

Why tests were silent

No test exercised a real property
string against the lifter's actual
AP set end-to-end.

Fix

PR #67: atom-matching gate returns
INCONCLUSIVE (unmatched_atoms
populated) instead of a false verdict.

Evidence

Real build: 58/58 checks INCONCLUSIVE
with unstripped atoms (100%, as designed).